Lab 6 practice: Laplace transforms (Zill §4.2, §4.3)

Prep for the Wed Oct 21 lab quiz. Closed book, about 10 minutes per problem. Problem 1 is forward transforms, problems 2 to 4 are inverse transforms, problem 5 is unit step functions, and problems 6 to 8 are initial value problems solved with Laplace transforms (problem 8 has a step-function forcing). Convolution, $t\,f(t)$, periodic functions, the Dirac delta and systems are left to Lab 7. Every answer was checked independently (`lab6_check.py`). Pair names are the ones in `reference/LaplaceTable.pdf`.

Transform pairs (table rows, $\mathcal{L}\{f\} = F(s)$, k and a constants):

Row in the table	$f(t)$	$F(s)$
1	1	$\dfrac{1}{s}$
t^n	t^n, $n = 0, 1, 2, \ldots$	$\dfrac{n!}{s^{n+1}}$
e^{at}	e^{at}	$\dfrac{1}{s-a}$
$\cos kt$	$\cos kt$	$\dfrac{s}{s^2+k^2}$
$\sin kt$	$\sin kt$	$\dfrac{k}{s^2+k^2}$
$e^{at}f(t)$ (first translation)	$e^{at}f(t)$	$F(s-a)$
$\dfrac{df}{dt}$	f'	$sF(s) - f(0)$
$\dfrac{d^2f}{dt^2}$	f''	$s^2F(s) - sf(0) - f'(0)$
$f(t-a)U(t-a)$ (second translation)	$f(t-a)U(t-a)$	$e^{-as}F(s)$
$g(t)U(t-a)$	$g(t)U(t-a)$	$e^{-as}\mathcal{L}\{g(t+a)\}$

All transforms are linear. Inverting: $\mathcal{L}^{-1}\{1/(s-a)^{n+1}\} = t^n e^{at}/n!$ (rows t^n and $e^{at}f(t)$ together), $\mathcal{L}^{-1}\{k/(s^2+k^2)\} = \sin kt$, so a numerator that is not k needs a factor $1/k$. For s^2+bs+c with complex roots, complete the square to $(s+p)^2+q^2$, rewrite the numerator in terms of $s+p$, and read off $e^{-pt}\cos qt$ and $e^{-pt}\sin qt$. For an IVP, transform both sides, solve for $Y(s)$, split into partial fractions, invert.

Problem 1

Find $\mathcal{L}\{f\}$ for each.

(a) $f(t) = (t+2)^2 + 4\sin t \cos t - 5e^{-2t}$

(b) $f(t) = e^{-3t}\left(t^2 - 2t + 4\right) + e^t\left(2\sin 3t - \cos 3t\right)$

(c) $f(t) = \sin^2 t$, using the transform of a derivative, then confirm with a trig identity.

Solution: forward transforms

(a) Expand and use a double-angle identity first, so that every piece is a table row: $(t+2)^2 = t^2+4t+4$ and $4\sin t\cos t = 2\sin 2t$. So $f(t) = t^2 + 4t + 4 + 2\sin 2t - 5e^{-2t}$. Row by row:

- t^n with $n = 2, 1, 0$: $\mathcal{L}\{t^2\} = 2/s^3$, $\mathcal{L}\{t\} = 1/s^2$, and $\mathcal{L}\{4\} = 4/s$ (row 1).
- $\sin kt$ with $k = 2$: $\mathcal{L}\{2\sin 2t\} = 2 \cdot 2/(s^2+4)$.
- e^{at} with $a = -2$: $\mathcal{L}\{-5e^{-2t}\} = -5/(s+2)$.

$$\boxed{F(s) = \frac{2}{s^3} + \frac{4}{s^2} + \frac{4}{s} + \frac{4}{s^2+4} - \frac{5}{s+2}}$$

(b) Each term is $e^{at}\times$ (something in the table), so use the first translation theorem, $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$.

First group, $a = -3$ and $f(t) = t^2 - 2t + 4$, with $F(s) = \dfrac{2}{s^3} - \dfrac{2}{s^2} + \dfrac{4}{s}$ (rows t^n and 1). Replace s by $s+3$:

$$\frac{2}{(s+3)^3} - \frac{2}{(s+3)^2} + \frac{4}{s+3}$$

Second group, $a = 1$ and $f(t) = 2\sin 3t - \cos 3t$, with $F(s) = \dfrac{6}{s^2+9} - \dfrac{s}{s^2+9}$ (rows $\sin kt$, $\cos kt$, $k = 3$). Replace s by $s-1$:

$$\frac{6-(s-1)}{(s-1)^2+9} = \frac{7-s}{(s-1)^2+9}$$

$$\boxed{F(s) = \frac{2}{(s+3)^3} - \frac{2}{(s+3)^2} + \frac{4}{s+3} + \frac{7-s}{(s-1)^2+9}}$$

(c) $f(0) = \sin^2 0 = 0$ and $f'(t) = 2\sin t\cos t = \sin 2t$. Row $\dfrac{df}{dt}$ gives $\mathcal{L}\{f'\} = sF(s) - f(0) = sF(s)$, and row $\sin kt$ $(k = 2)$ gives $\mathcal{L}\{\sin 2t\} = 2/(s^2+4)$. So $sF(s) = 2/(s^2+4)$.

$$\boxed{F(s) = \frac{2}{s\,(s^2+4)}}$$

Check with $\sin^2 t = \frac{1}{2}(1-\cos 2t)$ (rows 1 and $\cos kt$): $\frac{1}{2}\Big(\dfrac{1}{s} - \dfrac{s}{s^2+4}\Big) = \dfrac{1}{2}\cdot\dfrac{(s^2+4)-s^2}{s(s^2+4)} = \dfrac{2}{s(s^2+4)}$. Same.

Problem 2

Find $\mathcal{L}^{-1}\left\{\dfrac{5s^2 - s + 4}{(s+1)(s^2+4)}\right\}$.

Solution: partial fractions, linear factor plus irreducible quadratic

$s^2 + 4$ has no real roots, so the decomposition needs a linear numerator over the quadratic:

$$\frac{5s^2 - s + 4}{(s+1)(s^2+4)} = \frac{A}{s+1} + \frac{Bs + C}{s^2 + 4}$$

Clear denominators: $5s^2 - s + 4 = A(s^2 + 4) + (Bs + C)(s + 1)$.

- $s = -1$: $5 + 1 + 4 = 10 = 5A$, so $A = 2$.
- s^2 coefficients: $5 = A + B$, so $B = 3$.
- Constants: $4 = 4A + C$, so $C = -4$.

(Check the s coefficients: $-1 = B + C = 3 - 4$. Good.)

$$\frac{2}{s+1} + \frac{3s}{s^2+4} - \frac{4}{s^2+4}$$

Invert with row e^{at} ($a = -1$), row $\cos kt$ ($k = 2$), and row $\sin kt$ ($k = 2$). The last term is $-4/(s^2+4) = -2 \cdot \dfrac{2}{s^2+4}$, so it inverts to $-2 \sin 2t$.

$$\boxed{f(t) = 2e^{-t} + 3\cos 2t - 2\sin 2t}$$

Problem 3

Find $\mathcal{L}^{-1}\left\{\dfrac{4s^2-15s+4}{s\,(s-2)^2}\right\}$.

Solution: partial fractions with a repeated factor

A repeated factor $(s-2)^2$ needs a term for each power up to 2:

$$\frac{4s^2-15s+4}{s(s-2)^2} = \frac{A}{s} + \frac{B}{s-2} + \frac{C}{(s-2)^2}$$

Clear denominators: $4s^2-15s+4 = A(s-2)^2 + Bs(s-2) + Cs$.

- $s=0$: $4 = 4A$, so $A = 1$.
- $s=2$: $16-30+4 = -10 = 2C$, so $C = -5$.
- s^2 coefficients: $4 = A + B$, so $B = 3$.

$$\frac{1}{s} + \frac{3}{s-2} - \frac{5}{(s-2)^2}$$

Invert: row 1 gives 1, row e^{at} with $a=2$ gives e^{2t}, and $\mathcal{L}^{-1}\{1/(s-2)^2\} = te^{2t}$ (row t^n with $n=1$, shifted by the first translation theorem, $a=2$).

$$\boxed{f(t) = 1 + 3e^{2t} - 5te^{2t}}$$

Problem 4

Find $\mathcal{L}^{-1}\left\{\dfrac{3s^2+15s+22}{(s-1)(s^2+6s+13)}\right\}$.

Solution: completing the square

$s^2 + 6s + 13$ has discriminant $36 - 52 < 0$, so it stays together:

$$\frac{3s^2 + 15s + 22}{(s-1)(s^2+6s+13)} = \frac{A}{s-1} + \frac{Bs + C}{s^2 + 6s + 13}$$

$3s^2 + 15s + 22 = A(s^2 + 6s + 13) + (Bs + C)(s - 1)$.

- $s = 1$: $3 + 15 + 22 = 40 = 20A$, so $A = 2$.
- s^2 coefficients: $3 = A + B$, so $B = 1$.
- Constants: $22 = 13A - C = 26 - C$, so $C = 4$.

Complete the square: $s^2 + 6s + 13 = (s + 3)^2 + 4$, so $p = 3$ and $q = 2$. Rewrite the numerator in terms of $s + 3$: $s + 4 = (s + 3) + 1$.

$$\frac{2}{s-1} + \frac{(s+3)+1}{(s+3)^2+2^2} = \frac{2}{s-1} + \frac{s+3}{(s+3)^2+2^2} + \frac{1}{2} \cdot \frac{2}{(s+3)^2+2^2}$$

Invert. Row e^{at} $(a = 1)$ gives $2e^t$. The other two are rows $\cos kt$ and $\sin kt$ with $k = 2$, with s replaced by $s + 3$, so the first translation theorem $(a = -3)$ multiplies each by e^{-3t}.

$$\boxed{f(t) = 2e^t + e^{-3t}\cos 2t + \tfrac{1}{2}e^{-3t}\sin 2t}$$

The $\frac{1}{2}$ is the usual slip: $\sin 2t$ transforms to $2/(\cdot)$, not $1/(\cdot)$, so a lone 1 in the numerator has to be written as $\frac{1}{2} \cdot 2$.

Problem 5

(a) Find $\mathcal{L}\{t^2\,U(t-1)\}$.

(b) Let $f(t) = \sin t$ for $0 \le t < \pi$ and $f(t) = 0$ for $t \ge \pi$. Write f with unit step functions and find $\mathcal{L}\{f\}$.

(c) Find $\mathcal{L}^{-1}\left\{\dfrac{s\,e^{-\pi s/2}}{s^2+4}\right\}$.

(d) Find $\mathcal{L}^{-1}\left\{\dfrac{e^{-2s}}{s\,(s+3)}\right\}$.

Solution: the second translation theorem both ways

(a) Here the function multiplying $U(t-1)$ is $g(t) = t^2$, not a shifted function, so use row $g(t)U(t-a)$ with $a = 1$: $e^{-s}\mathcal{L}\{g(t+1)\}$. $g(t+1) = (t+1)^2 = t^2 + 2t + 1$, and with rows t^n and 1:

$$\boxed{\mathcal{L}\{t^2U(t-1)\} = e^{-s}\left(\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s}\right)}$$

(b) The function switches off at $t = \pi$: $f(t) = \sin t - \sin t\, U(t-\pi)$. To use row $f(t-a)U(t-a)$ the sine must be written in terms of $t - \pi$, and $\sin t = -\sin(t-\pi)$. So

$$f(t) = \sin t + \sin(t-\pi)\, U(t-\pi)$$

With rows $\sin kt$ $(k = 1)$ and $f(t-a)U(t-a)$ $(a = \pi)$:

$$\boxed{\mathcal{L}\{f\} = \frac{1}{s^2+1} + \frac{e^{-\pi s}}{s^2+1} = \frac{1+e^{-\pi s}}{s^2+1}}$$

Check with row $g(t)U(t-a)$ instead: $g(t) = \sin t$, $g(t+\pi) = -\sin t$, so the second term is $-\sin t\, U(t-\pi) \mapsto e^{-\pi s}\cdot(-\frac{1}{s^2+1})$, and subtracting it gives $+\dfrac{e^{-\pi s}}{s^2+1}$. Same.

(c) The factor $e^{-\pi s/2}$ says $a = \pi/2$. Without it, $F(s) = s/(s^2+4)$, which is row $\cos kt$ with $k = 2$, so $f(t) = \cos 2t$. By row $f(t-a)U(t-a)$:

$$\mathcal{L}^{-1}\{e^{-\pi s/2}F(s)\} = \cos\left(2(t-\tfrac{\pi}{2})\right)U(t-\tfrac{\pi}{2})$$

Since $\cos(2t-\pi) = -\cos 2t$:

$$\boxed{f(t) = -\cos 2t\; U(t-\tfrac{\pi}{2}) = \begin{cases} 0, & 0 \le t < \pi/2 \\ -\cos 2t, & t \ge \pi/2 \end{cases}}$$

(d) Partial fractions on the part without the exponential: $\dfrac{1}{s(s+3)} = \dfrac{1}{3}\Big(\dfrac{1}{s} - \dfrac{1}{s+3}\Big)$. Rows 1 and e^{at} $(a = -3)$ give $f_0(t) = \frac{1}{3}(1 - e^{-3t})$. With $a = 2$, row $f(t-a)U(t-a)$ replaces t by $t-2$ everywhere, including inside the exponential:

$$\boxed{f(t) = \tfrac{1}{3}\Big(1 - e^{-3(t-2)}\Big)\,U(t-2) = \begin{cases} 0, & 0 \le t < 2 \\ \frac{1}{3}(1 - e^{-3(t-2)}), & t \ge 2 \end{cases}}$$

Problem 6

Solve $y'' + 2y' + 5y = 8e^{-t}, \quad y(0) = 2, \quad y'(0) = -4.$

Solution: initial value problem, complex roots

Second order, linear, constant coefficients, with initial values given at $t = 0$, so Laplace transforms fit. Let $Y(s) = \mathcal{L}\{y\}$. Rows $\dfrac{d^2f}{dt^2}$, $\dfrac{df}{dt}$ and e^{at} $(a = -1)$:

$$\big[s^2Y - 2s + 4\big] + 2\big[sY - 2\big] + 5Y = \frac{8}{s+1}$$

$$(s^2 + 2s + 5)\,Y = 2s + \frac{8}{s+1}$$

The constants $+4$ and -4 cancelled. Combine and solve:

$$Y = \frac{2s(s+1) + 8}{(s+1)(s^2+2s+5)} = \frac{2s^2 + 2s + 8}{(s+1)(s^2+2s+5)} = \frac{A}{s+1} + \frac{Bs + C}{s^2 + 2s + 5}$$

$2s^2 + 2s + 8 = A(s^2 + 2s + 5) + (Bs + C)(s + 1)$.

- $s = -1$: $2 - 2 + 8 = 8 = 4A$, so $A = 2$.
- s^2 coefficients: $2 = A + B$, so $B = 0$.
- Constants: $8 = 5A + C = 10 + C$, so $C = -2$.

$s^2 + 2s + 5 = (s + 1)^2 + 2^2$, so

$$Y = \frac{2}{s+1} - \frac{2}{(s+1)^2 + 2^2}$$

The second term is already $k/(\cdot)$ with $k = 2$. Row e^{at} gives $2e^{-t}$, and row $\sin kt$ with the first translation theorem $(a = -1)$ gives $-e^{-t}\sin 2t$.

$$\boxed{y(t) = 2e^{-t} - e^{-t}\sin 2t = e^{-t}\,(2 - \sin 2t)}$$

Check the initial values: $y(0) = 2$. $y' = -2e^{-t} + e^{-t}\sin 2t - 2e^{-t}\cos 2t$, so $y'(0) = -2 - 2 = -4$. Both match.

Problem 7

Solve $y'' - 6y' + 9y = t^2 e^{3t}, \quad y(0) = 1, \quad y'(0) = 5.$

Solution: initial value problem with a repeated root, forcing needs the first translation theorem

Let $Y = \mathcal{L}\{y\}$. The forcing t^2e^{3t} is $e^{at}f(t)$ with $a = 3$ and $f = t^2$, so its transform is $2/(s-3)^3$ (rows t^n and $e^{at}f(t)$). Transform the equation with rows $\dfrac{d^2f}{dt^2}$ and $\dfrac{df}{dt}$:

$$\big[s^2Y - s - 5\big] - 6\big[sY - 1\big] + 9Y = \frac{2}{(s-3)^3}$$

$$(s-3)^2\,Y = s - 1 + \frac{2}{(s-3)^3}$$

since $s^2 - 6s + 9 = (s-3)^2$, and moving the constants across gives $s + 5 - 6 = s - 1$.

$$Y = \frac{s-1}{(s-3)^2} + \frac{2}{(s-3)^5}$$

Rewrite $s - 1 = (s-3) + 2$:

$$Y = \frac{1}{s-3} + \frac{2}{(s-3)^2} + \frac{2}{(s-3)^5}$$

Invert with $\mathcal{L}^{-1}\{1/(s-3)^{n+1}\} = t^ne^{3t}/n!$: e^{3t}, then $2te^{3t}$, then $2 \cdot t^4e^{3t}/4! = \frac{1}{12}t^4e^{3t}$.

$$\boxed{y(t) = e^{3t}\Big(1 + 2t + \tfrac{1}{12}t^4\Big)}$$

Check: $y(0) = 1$. $y' = 3e^{3t}(1 + 2t + \frac{t^4}{12}) + e^{3t}(2 + \frac{t^3}{3})$, so $y'(0) = 3 + 2 = 5$.

Problem 8

Solve $y'' + 3y' + 2y = f(t), \quad y(0) = 0, \quad y'(0) = 1$, where $f(t) = 6$ for $1 \leq t < 3$ and $f(t) = 0$ otherwise.

Solution: initial value problem with a pulse forcing

Forcing. The pulse is on from $t = 1$ to $t = 3$: $f(t) = 6\,U(t-1) - 6\,U(t-3)$. Row $f(t-a)U(t-a)$ with $f = 6$ (row 1) gives

$$\mathcal{L}\{f\} = \frac{6\,(e^{-s} - e^{-3s})}{s}$$

Transform the equation (rows $\dfrac{d^2f}{dt^2}$ and $\dfrac{df}{dt}$), with $Y = \mathcal{L}\{y\}$:

$$\big[s^2Y - 0 - 1\big] + 3\big[sY - 0\big] + 2Y = \frac{6\,(e^{-s} - e^{-3s})}{s}$$

$$(s+1)(s+2)\,Y = 1 + \frac{6\,(e^{-s} - e^{-3s})}{s}$$

$$Y = \frac{1}{(s+1)(s+2)} + 6\,(e^{-s} - e^{-3s})\,\frac{1}{s(s+1)(s+2)}$$

Partial fractions. $\dfrac{1}{(s+1)(s+2)} = \dfrac{1}{s+1} - \dfrac{1}{s+2}$, and $\dfrac{1}{s(s+1)(s+2)} = \dfrac{1}{2s} - \dfrac{1}{s+1} + \dfrac{1}{2(s+2)}$ (cover-up: the values at $s = 0, -1, -2$ are $\frac{1}{2}$, -1, $\frac{1}{2}$). Multiplying the second by 6 and inverting rows 1 and e^{at} term by term gives

$$h(t) = \mathcal{L}^{-1}\left\{\frac{6}{s(s+1)(s+2)}\right\} = 3 - 6e^{-t} + 3e^{-2t}$$

The first part inverts to $e^{-t} - e^{-2t}$. The exponentials e^{-s} and e^{-3s} shift h (row $f(t-a)U(t-a)$, $a = 1$ and $a = 3$):

$$\boxed{y(t) = e^{-t} - e^{-2t} + h(t-1)\,U(t-1) - h(t-3)\,U(t-3), \qquad h(t) = 3 - 6e^{-t} + 3e^{-2t}}$$

Piecewise:

$$y(t) = \begin{cases} e^{-t} - e^{-2t}, & 0 \le t < 1 \\ e^{-t} - e^{-2t} + 3 - 6e^{-(t-1)} + 3e^{-2(t-1)}, & 1 \le t < 3 \\ e^{-t} - e^{-2t} - 6e^{-(t-1)} + 6e^{-(t-3)} + 3e^{-2(t-1)} - 3e^{-2(t-3)}, & t \ge 3 \end{cases}$$

Sense check: for $0 \le t < 1$ there is no forcing, so y is the free response with the given $y(0) = 0$, $y'(0) = 1$, and indeed $y'(0) = -1 + 2 = 1$. After $t = 3$ the forcing is off again and y decays, as the last line does. Because $h(0) = 0$ and $h'(0) = 0$, the pieces join with y and y' continuous at $t = 1$ and $t = 3$, which is what a bounded forcing should give for a second-order equation.